

PROFILE

Product-focused full-stack engineer with 2+ years of full-time experience shipping React, TypeScript, Node.js, AWS, and applied-AI workflows from ambiguous ideas through tested delivery.

TECHNICAL SKILLS

Product & Frontend: TypeScript, JavaScript, React, responsive interfaces, workflow design

Backend & Applied AI: Node.js, Express, Python, FastAPI, REST APIs, AWS S3, RAG, vector search

Reliability & Quality: pytest, Vitest, Testing Library, Playwright, security rules, workflow validation

EXPERIENCE

Founder and Lead Developer | Tallyrus

Jun 2023-Present

Full-stack retrieval tooling for educator feedback

Python | React | TypeScript | Flask | MongoDB | LangChain

- Built an educator-facing AI product across **4 connected workflows** - rubric setup, document ingestion, retrieval-grounded scoring, and feedback review - using React, TypeScript, Python, and Pinecone to turn an ambiguous grading problem into a testable end-to-end experience for structured educator review.
- Implemented **5 application safeguards** across Node.js and React services - Google OAuth, protected routes, secure sessions, rate limiting, and Helmet headers - while MongoDB and S3 preserved application and document state for reliable educator workflows.
- Led **3 product-delivery loops** spanning user interviews, rubric iteration, and full-stack releases, translating educator feedback into scoped features and helping secure **five figures** for continued development through validated product decisions and review workflows.
- Connected **3 persistence systems** - MongoDB application state, S3 documents, and Pinecone retrieval - through Python and Node.js services so rubric edits, uploaded evidence, and generated feedback stayed synchronized across the educator workflow.
- Owned **4 delivery functions** - product planning, user testing, full-stack implementation, and fundraising - turning educator feedback into scoped releases, secure review paths, documented product decisions, and continued early-stage product development.

Business Technology Solutions Associate | ZS Associates

South San Francisco, CA | Jun 2024-Present

Healthcare AI, data, API integration, and workflow delivery

Python | FastAPI | React | Snowflake | AWS | pytest

- Developed Snowflake SQL pipelines for master-data extraction, identifier matching, and API connectivity across **327K+ healthcare records**, producing auditable quality-control reports and exception outputs for repeatable regulated stakeholder review, exception resolution, audit, and stewardship decisions.
- Connected **3 delivery surfaces** - Python services, AWS S3 storage, and Smartsheet workflows - through Flask endpoints and documented UAT evidence, automating document handoffs while keeping implementation decisions and stakeholder validation traceable.
- Built React and FastAPI knowledge-search services connecting documents and user questions across **2 backend layers** - AWS data services and Redis retrieval state - so teams could evaluate grounded evidence with less manual coordination and repeatable service behavior.

Data Analyst / Web Development Lead | Tech4Good @ UCSC

Mar 2023-Present

Collaborative research and classroom workflow tooling

TypeScript | Node.js | Express | MongoDB | Python | Figma APIs

- Contributed **158 GitHub commits**, including **44 backend-scoped commits** across TypeScript and Figma API chat, polling, Likert, text-entry, and export workflows in a collaborator-owned research toolkit, preserving accurate team ownership, public attribution, and research delivery evidence.
- Implemented **20+ Git-attributed Node/Express REST routes** backed by three MongoDB databases, populated message logs, change streams, and persisted poll and voter records, reliably synchronizing threads, edits, anonymous responses, votes, grouped responses, reveal operations, and live updates.
- Analyzed **43,340 anonymized interaction rows** across **2,805 session IDs** with Python and Pandas notebooks, while six team-published Figma Community resources reached approximately **12.2K aggregate uses** with team-library attribution preserved for reproducible analytics and public research reporting.

PROJECTS

Sylk | Multi-Tenant Supply-Chain Platform

2025-Present

Role-aware web and Android product with tested tenant isolation

React 19 | TypeScript | Firebase | Capacitor 7 | Playwright

- Shipped **6 customer workflows** for organization onboarding, shipments, fulfillment, tracking, file collaboration, and real-time chat using React 19 and TypeScript, separating retailer financial data from shared factory operations across responsive interfaces.
- Protected **3 user groups** with Firebase Authentication plus Firestore and Storage rules, then exercised cross-tenant permissions and complete product journeys through Vitest, **Testing Library**, emulators, and Playwright before packaging responsive web and Android releases.
- Delivered **1 Android product package** with Capacitor 7, native-safe routing, Firebase Authentication, file picking, and Gradle automation, validating a 7.8 MB debug APK and minimum SDK 23 across repeatable build, compatibility, authentication, and mobile file-handling checks.

Lume | AI Meeting Intelligence Platform

Dec 2025-Present

Document-grounded meeting knowledge and search

React 18 | Node.js | MongoDB | AWS S3 | WebSockets | RAG

- Built **4 intelligent meeting workflows** for capture, storage, search, and grounded question answering, connecting React to Node.js, MongoDB, AWS S3, and WebSockets so users could move from raw conversations to reusable project context and citation-backed evidence across repeated sessions.
- Designed a **5-stage retrieval-augmented generation flow** using vector embeddings, intent routing, evidence extraction, citation-backed answers, and generator/critic refinement, keeping responses grounded in meeting and document context with human-review evidence instead of unsupported model output.
- Connected **2 knowledge sources** - meeting transcripts and seed files - into reusable Project Brain knowledge bases preserving terminology, relationships, and project context for transcript-aware search, citation-backed document answers, reusable user capabilities, and repeated knowledge workflows across sessions.

Private Filedrop | AI-Integrated File Exchange

2026

Secure APIs and cross-session AI coordination

Node.js | Express | AWS S3 | OpenAPI | MCP | HMAC

- Secured **2 access paths** with password-protected browser uploads and API-key-protected programmatic exchange, adding upload validation, metadata, categories, tags, project paths, auditable version history, and unguessable direct-download links for controlled secure cross-session file handoff systems.
- Implemented **2 storage drivers** - local and S3-compatible - plus documented migration tooling for durable deployment, while REST, OpenAPI, and MCP interfaces exposed upload, download, organization, versioning, message, channel, and webhook operations to browsers, APIs, and AI clients.
- Protected automation using **2 webhook verification controls** with HMAC-SHA256 signatures and timing-safe checks, then added shared mailbox and channel abstractions so sessions could exchange control messages and events with cross-session auditability without exposing storage credentials.
- Documented **4 supported operating modes** - local use, cloud deployment, ChatGPT Actions, and MCP clients - after deploying through Render configuration, giving browser users, API consumers, and AI sessions repeatable setup paths for secure file storage, messaging, and cross-session coordination.

EDUCATION

University of California, Davis | B.S. Computer Science | GPA: 3.7

Davis, CA | Graduated Jun 2025

Relevant Foundation: Algorithms, Databases, Distributed Systems, Operating Systems, AI/ML, Data Structures

PUBLICATION

ICER 2024 - Evaluating Exploratory Reading Groups for Supporting Undergraduate Research Pipelines in Computing - Research examining how structured exploratory reading groups support undergraduate computing research pipelines.

LINKS

Portfolio: <https://ashchembu.com> | LinkedIn: [linkedin.com/in/ashwin-chembu-a5292b205](https://www.linkedin.com/in/ashwin-chembu-a5292b205) | GitHub: github.com/ashwinchembu